AI Capital versus Regular Capital: Output, Real Interest Rates, Wages and the Labor Share after a Jump, an S-Curve, or Both

Research note (draft)

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Abstract

This note compares how output, the real interest rate, the real wage and the labor share respond to technical progress in AI capital and in regular capital. Regular capital and labor form a Cobb-Douglas bundle, and AI capital substitutes for that bundle ($\varepsilon = 1.5$). Progress is either disembodied (in the production function) or embodied (a lower price of new capital in the accumulation equation). Both capital stocks face the same moderate adjustment costs. Three timing patterns are compared: a surprise permanent jump; an S-curve centered on year 10 and known at year 0; and a mix in which half the improvement arrives as a jump and half along the S-curve. Elasticities are the same throughout, and every shock raises long-run output by the same 10 percent. In the long run, which is in closed form, AI progress raises the wage by 7.4 percent and lowers the labor share by 1.8 points. Regular-capital progress raises the wage by the full 10 percent and leaves the labor share unchanged. In the short run, the interest rate rises most for embodied AI progress after a jump, and most for disembodied regular-capital progress along an S-curve. Output and wage gains arrive fastest under disembodied progress. Output converges faster for AI than for regular capital, but AI wages lag AI output. The mixed case lies close to the average of the two pure cases.

1 Model

Notation follows Kiley (2000). Production is

$$X = (\Gamma_K K)^a (ZN)^{1-a}, \qquad Y = \left[\omega^{1/\varepsilon} X^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)^{1/\varepsilon} (\Gamma_A A)^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}}, \qquad \varepsilon = 1.5,$$

so AI capital A substitutes for both labor and regular capital K. Capital is measured in efficiency units and accumulates as $K_j' = (1-d_j)K_j + I_j/P_j$, where P_j is the output price of a new efficiency unit. Adjustment costs, following Kiley (2004), are

$$P_j \, \tfrac{c_j}{2} \, (i_j - x_j)^2 K_j, \qquad i_j = \frac{I_j}{P_j K_j}, \qquad c_j = \frac{1}{x_j}.$$

They vanish at the balanced-growth investment rate $x_j = g + d_j$, and they give both stocks the same unit elasticity of investment with respect to Q_j/P_j (the paper's "moderate" case). *Disembodied* progress raises Γ_j, which improves the installed stock. *Embodied* progress lowers P_j, which improves only new investment. Utility is logarithmic and hours are fixed.

Calibration. $a = 0.282$, $d_K = 0.066$, $d_A = 0.20$, trend growth $g = 1.5$ percent, initial $r = 7.5$ percent, initial AI income share 3 percent, initial labor share 69.6 percent.

Proposition 1 (Balanced growth path)**.** *With constant technology levels, $Q_j = P_j$, $1+r = (1+g)/B$ and $u_j = r + d_j$. Let $\lambda_j = P_j u_j/\Gamma_j$ be the cost of an effective unit of capital j. Then*

$$s_A = (1-\omega)\lambda_A^{1-\varepsilon}, \qquad \textit{labor share} = (1-a)(1-s_A), \qquad P_X = \Big(\frac{1-s_A}{\omega}\Big)^{\frac{1}{1-\varepsilon}},$$
$$\ln\frac{W}{Z} = \frac{\ln P_X}{1-a} - \frac{a}{1-a}\ln\lambda_K + \textit{const}, \qquad \ln\frac{Y}{Z} = \ln\frac{W}{Z} - \ln(1-s_A) + \textit{const}.$$

Proposition 2 (Long run, same output gain γ_Y)**.** *Disembodied and embodied progress have the same long-run effects ($\ln\Gamma_j = -\ln P_j$), and the interest rate returns to its initial level.*

1. Regular capital*: $\ln(\Gamma_K/P_K) = \frac{1-a}{a}\gamma_Y$, $\Delta\ln W = \gamma_Y$, and the labor share is unchanged.*
2. AI capital*: $\Delta\ln W = \dfrac{\gamma_Y}{1+(1-a)(\varepsilon-1)}$ and $\Delta\ln(\textit{labor share}) = -\dfrac{(1-a)(\varepsilon-1)}{1+(1-a)(\varepsilon-1)}\gamma_Y$.*

Proof. By Proposition 1, $\ln(Y/Z) = \big[\frac{1}{(1-a)(1-\varepsilon)} - 1\big]\ln(1-s_A) - \frac{a}{1-a}\ln\lambda_K + \text{const}$, and $\ln W = \ln Y + \ln(1-s_A)$. Regular-capital progress moves only λ_K, and AI progress moves only s_A. □

For $\gamma_Y = 10$ percent, regular-capital technology rises 29 percent, against 3.4-fold for AI, because AI starts small. AI's income share rises from 3.0 to 5.5 percent. In the long run, AI progress raises the wage 7.4 percent and lowers the labor share from 69.6 to 67.8 percent. Regular-capital progress raises the wage 10 percent and leaves the labor share unchanged.

2 Results

All experiments start on the initial balanced growth path. In the *jump*, technology rises permanently and unexpectedly at $t = 0$. In the *S-curve*, a logistic rise centered on year 10 is announced at $t = 0$. In the *half/half* case, half the log improvement arrives as a jump at $t = 0$ and the other half along the same S-curve, also announced at $t = 0$. Each is solved as a nonlinear perfect-foresight problem. Output (net of adjustment costs) and wages are log deviations from the pre-shock path, and the labor share is relative to gross output.

Output

- *Jump.* Disembodied progress raises output on impact, by 6.9 percent for regular capital and 4.1 for AI. Embodied progress cannot raise output until new capital is installed, and net output dips on impact (by 0.4 percent for regular capital and 1.5 for AI) because the investment boom incurs adjustment costs. AI output then catches up quickly. AI capital depreciates fast and has low adjustment costs per unit of investment, so after 10 years output is up 7.8 (AI, disembodied) and 6.6 (AI, embodied), against 8.3 and 4.3 for regular capital.
- *S-curve.* Output tracks the innovation. At year 20 it is up 8.1 (regular, disembodied), 7.8 (AI, disembodied), 5.8 (AI, embodied) and 3.8 percent (regular, embodied). Embodied regular-capital progress is the slowest to raise output in every timing pattern.
- *Half/half.* Output jumps by about half the pure-jump amount and then follows the S-curve.

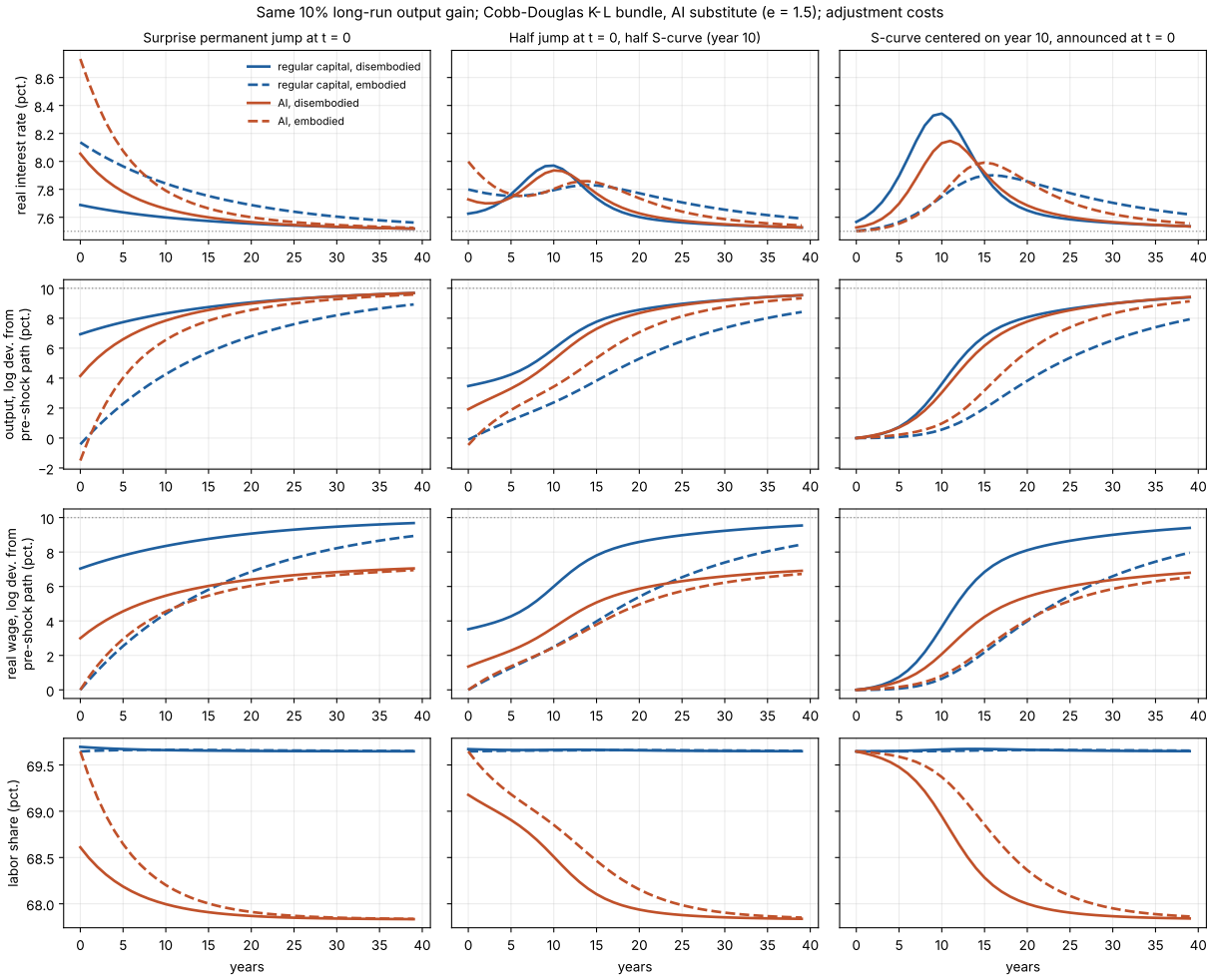


Figure 1: Same 10% long-run output gain in every experiment. Dotted lines: initial r and long-run output gain.

Real interest rate

- *Jump.* The rate jumps at once and then declines. On impact the ranking is AI embodied (8.7 percent), regular embodied (8.1), AI disembodied (8.05) and regular disembodied (7.7). Embodied progress raises r more because the output gain requires an investment boom first. AI raises r more than regular capital because a larger share of its long-run gain must be accumulated. Disembodied AI delivers 3.0 of its 7.4 percent wage gain on impact, against 7.0 of 10 percent for regular capital. Over 40 years, the excess return adds up to about 9 percentage points in both embodied cases, against 4.7 for disembodied AI and 2.7 for disembodied regular capital.

- *S-curve.* The rate barely moves at announcement. Under disembodied progress it rises ahead of the innovation and peaks with it (year 10–11): 8.3 percent for regular capital and 8.15 for AI. Under embodied progress it peaks later (years 15–16) and lower, at 7.9 for regular capital and 8.0 for AI, but the response lasts longer. The 40-year cumulative effects are similar, 7.4–9.4 percentage points.

Table 1: Summary. r_1 is the real rate between years 0 and 1. All shocks raise long-run output by 10%.

		real rate (pct.)			output, $\ln Y$ (pct.)			real wage, $\ln W$ (pct.)			labor share
		r_1	peak	year	t=0	t=10	t=20	t=0	t=10	t=20	t=10 (pct.)
Jump	Regular, disemb.	7.69	7.69	0	6.9	8.3	9.1	7.0	8.4	9.1	69.7
	Regular, emb.	8.14	8.14	0	-0.4	4.3	6.8	0.0	4.4	6.9	69.7
	AI, disemb.	8.05	8.05	0	4.1	7.8	9.0	3.0	5.5	6.4	68.0
	AI, emb.	8.73	8.73	0	-1.5	6.6	8.6	0.0	4.5	6.0	68.2
Half/half	Regular, disemb.	7.63	7.97	10	3.5	6.0	8.6	3.5	6.0	8.6	69.7
	Regular, emb.	7.80	7.83	14	-0.1	2.4	5.3	0.0	2.5	5.4	69.7
	AI, disemb.	7.73	7.94	10	1.9	5.2	8.3	1.4	3.6	5.9	68.5
	AI, emb.	8.00	8.00	0	-0.5	3.4	7.1	0.0	2.5	5.0	68.9
S-curve	Regular, disemb.	7.57	8.34	10	0.0	3.6	8.1	0.0	3.7	8.1	69.7
	Regular, emb.	7.50	7.90	16	0.0	0.6	3.8	0.0	0.6	4.0	69.7
	AI, disemb.	7.53	8.15	11	0.0	3.0	7.8	0.0	2.1	5.4	68.9
	AI, emb.	7.50	7.99	15	0.0	1.0	5.8	0.0	0.8	4.1	69.4

- *Half/half.* The rate has two humps, an impact response from the jump and a second peak as the S-curve arrives (years 10–14). For disembodied progress the second peak is larger (7.97 regular, 7.94 AI). For embodied AI the impact response is larger (8.00). Each path is close to the average of the pure jump and pure S-curve paths. For example, $r_1 = 7.63$ for disembodied regular capital, against an average of 7.63.

Real wage

- *Long run*: 10.0 percent for regular capital and 7.4 percent for AI (Proposition 2).
- *Jump.* Disembodied progress raises the wage on impact, by 7.0 percent for regular capital and 3.0 for AI. Embodied progress starts from zero. By year 10 the wage is up 8.4 (regular, disembodied), 5.5 (AI, disembodied), 4.5 (AI, embodied) and 4.4 percent (regular, embodied).
- *S-curve.* The wage follows the innovation, and much more slowly when progress is embodied. At year 10 it is up 3.7 percent (regular, disembodied), 2.1 (AI, disembodied), 0.8 (AI, embodied) and 0.6 (regular, embodied). At year 20 the figures are 8.1, 5.4, 4.1 and 4.0.
- *Half/half.* Wages jump by half the disembodied pure-jump amount (3.5 regular, 1.4 AI). By year 10 they are up 6.0 (regular, disembodied), 3.6 (AI, disembodied), 2.5 (AI, embodied) and 2.5 percent (regular, embodied).
- *AI wages lag AI output.* Because the AI share is rising, wages grow more slowly than output under AI progress. At year 10 of the jump, AI output is up 7.8 percent but wages only 5.5 percent. Under regular-capital progress the two move together.

Labor share

- *Regular capital*: essentially unchanged throughout, within 0.05 points.
- *AI*: falls 1.8 points to 67.8 percent. After a disembodied jump, about 60 percent of the fall happens on impact. After an embodied jump it happens as AI capital is installed (68.2 percent at year 10). Along the S-curve it tracks the innovation, at 68.9 percent (disembodied)

or 69.4 percent (embodied) in year 10. In the half/half case, half the disembodied fall happens on impact (to 69.2 percent).

3 Summary

With the same elasticities and the same long-run output gain:

1. AI progress delivers about three-quarters of the wage gain of regular-capital progress and lowers the labor share. Regular-capital progress leaves the labor share unchanged. These long-run results are in closed form and do not depend on whether progress is embodied or disembodied.

2. Output gains arrive fastest under disembodied progress. Under embodied AI progress they arrive faster than under embodied regular-capital progress, because AI capital turns over quickly. Embodied jumps briefly lower net output while the investment boom absorbs resources.

3. The interest rate rises temporarily in every case. A surprise jump raises it most when progress is embodied and in AI. An anticipated S-curve raises it most, and soonest, when progress is disembodied. Embodied progress produces a later, smaller but more persistent rise.

4. AI wages lag AI output, and embodied progress delays wage gains by roughly five to ten years compared with disembodied progress.

5. A mix of a jump and an S-curve gives approximately the average of the two pure responses, so the timing patterns can be combined roughly linearly.

A Verification

`capital_cases.py` produces all results. It checks that the closed-form balanced growth path satisfies the equilibrium conditions, that every experiment (jump, half/half, S-curve) converges to the 10 percent output gain, and that long-run wages match Proposition 2. Newton residuals are below 10^{-10}.

References

Kiley, M. T. (2000). "Stock Prices and Fundamentals in a Production Economy." FEDS Working Paper 2000-05, Federal Reserve Board.

Kiley, M. T. (2004). "Stock Prices and Fundamentals: A Macroeconomic Perspective." *Journal of Business* 77(4): 909–936.

Check	max abs. error
closed-form BGP satisfies equilibrium	1.3e-15
jump K dis: Newton residual	2.7e-12
jump K dis: long-run ln Y gain vs 10jump K emb: Newton residual	8.5e-13
jump K emb: long-run ln Y gain vs 10jump A dis: Newton residual	3.0e-15
jump A dis: long-run ln Y gain vs 10jump A emb: Newton residual	3.8e-13
jump A emb: long-run ln Y gain vs 10mixed K dis: Newton residual	4.1e-12
mixed K dis: long-run ln Y gain vs 10mixed K emb: Newton residual	2.8e-13
mixed K emb: long-run ln Y gain vs 10mixed A dis: Newton residual	2.0e-15
mixed A dis: long-run ln Y gain vs 10mixed A emb: Newton residual	1.8e-15
mixed A emb: long-run ln Y gain vs 10scurve K dis: Newton residual	3.9e-12
scurve K dis: long-run ln Y gain vs 10scurve K emb: Newton residual	2.9e-13
scurve K emb: long-run ln Y gain vs 10scurve A dis: Newton residual	3.4e-11
scurve A dis: long-run ln Y gain vs 10scurve A emb: Newton residual	2.1e-15
scurve A emb: long-run ln Y gain vs 10jump K dis: long-run ln W vs closed form	3.4e-08
jump K emb: long-run ln W vs closed form	1.1e-07
mixed K dis: long-run ln W vs closed form	4.8e-08
mixed K emb: long-run ln W vs closed form	1.6e-07
scurve K dis: long-run ln W vs closed form	6.3e-08
scurve K emb: long-run ln W vs closed form	2.1e-07
jump A dis: long-run ln W vs closed form	3.4e-08
jump A emb: long-run ln W vs closed form	4.5e-08
mixed A dis: long-run ln W vs closed form	4.9e-08
mixed A emb: long-run ln W vs closed form	6.7e-08
scurve A dis: long-run ln W vs closed form	6.1e-08
scurve A emb: long-run ln W vs closed form	8.5e-08